LOKMANYA TILAK JANKALYAN SHIKSHAN SANSTHA'S

PRIYADARSHINI COLLEGE OF ENGINEERING

PRIYADARSHINI COLLEGE OF ENGG. NAGPUR

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

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Idea Submission Template | Code. Create. Innovate.

Team Name	**Team Raksha**
Problem Statement No.	**2**
Problem Statement Title	**Improving Safety at Large Public Events**
Team Leader Name	**Shounak Joshi**

THE PROBLEM

Guards watch screens. Danger builds unseen.

A control room watches dozens of flat camera feeds. A guard spots trouble only after the crowd stops moving. By then, crushing pressure is already passing body to body. Stillness is not the warning. It is the accident.

Hathras 2024
121 dead. 80k permit, 250k crowd. Exits blocked.

Mumbai 2017
23 dead on a station footbridge. Rain, rush hour, no metering.

7 per m2
A crowd turns fluid. Shockwaves throw people metres. (Fruin)

Kumbh, 29 January 2025

Months before
2,751 cameras and AI alerts go live (TOI).

01:00
Barricades jumped at the Sangam nose. Crores press forward.

After
30 dead officially, despite live AI alerts. No action followed in time.



Kumbh mela ground at night, Jan 2025. Photo: Wikimedia Commons.

Alerts existed. Action did not follow in time.

Why stillness lies
A stopped crowd is not a warning. It is the accident. Pressure passes body to body long before feet stop. That is why cameras must read force, not faces.

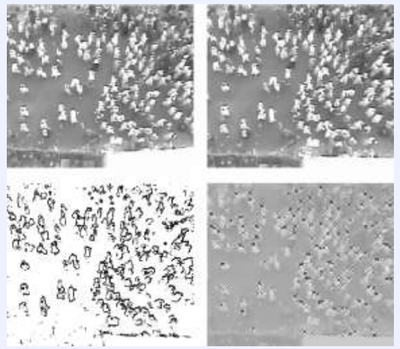
Danger is invisible pressure, not visible stillness.
Measure pressure, warn early, open gates.

HOW IT WORKS

Calibrate	Map density	Track flow	Measure pressure	Forecast jam	Warn
pixels to ground, once per camera	heatmaps count overlap	flow reads crowd currents	density x movement variance	inflow beats outflow	open gates upstream

In plain words: map the ground right, count the crowd as heat, read its currents, warn on pressure.

What the machine sees



Head-density surface: each peak is one head, the sum is the count. (Johansson 2008).



Flow field on our station clip (DIS optical flow). Movers glow, stillness stays black.

Heatmaps beat boxes

Published research shows box detectors collapse under overlap: F1 0.07 vs 0.71 point-based. Density maps cut counting error 265 to 105 MAE. (NWPU-Crowd)

Pressure, not stillness

Pressure is density times movement variance. Density or speed alone cannot warn. (Johansson 2008)

Warns before stillness

Stop-and-go waves and rising pressure fire the alert while feet still move. Shown on Hajj 2006 footage; multi-site trials are the pilot's job.

FEASIBILITY AND VIABILITY

One camera, one edge box

Camera: fixed gate view

↓

Edge box: density plus flow plus pressure on device

↓

Sends JSON numbers, not continuous video

↓

Venue servers (paired): fuse every camera into one map

↓

Guard wall: ranked zones, early warning

Compute target: 5 to 10 FPS per camera at reduced resolution; pressure moves over seconds, not frames. About $250 and 15 watts per edge box (1 to 4 cameras); server, handsets, network extra. Reuses existing CCTV.

Video stays put

Each edge box keeps a 10-minute rolling buffer and sends JSON numbers over site LAN or 4G failover. Only a confirmed Red alert exports its clip, signed.

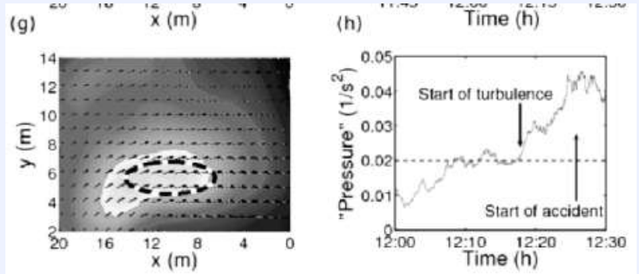
Calibrated where it counts

Fixed gates are measured once; Kumbh ran 3-month surveys for its alarm. Open grounds and PTZ cameras run relative levels plus entry counts.

Private by design

Heatmaps cannot show a face. Saved video is blurred, wiped after 7 days, zero identity in core. Police clips are access-logged, designed to align with DPDP Act 2023 principles.

Pressure warned first



Pressure (density x movement variance) crossed its warning line before turbulence began. The accident came after. Hajj 2006 data (Johansson et al.).

IMPACT AND BENEFITS

When pressure rises

Yellow watch
Auto-pages the zone guard with a 2-minute ack timer. Overflow gates get ready.

Red alert
Guards confirm on handsets; unanswered reds climb to venue command, then city control (planned link).

Handover
A sealed evidence pack goes to the police.

Pilot targets: warn before critical, alerts reach guards in seconds, publish the false-alarm rate. Still to prove: thresholds, survey-free calibration, night/rain/angle tests.

Before
A wall of screens, one tired guard, and risks get missed.

After
Pressure at Gate B opens overflow at Gate A upstream. Kumbh's alarms reportedly went unactioned; ours forces an answer: owner, timer, log.

Organizers
Spot a growing jam early, and tell guards which zone to clear. Built for stations, mela grounds, stadiums.

Police
Every alert, zone and action exports as one evidence pack in a click, with the signed 10-minute incident clip.

Public
Faces stay blurred in saved video.

RESEARCH AND REFERENCES

The science behind it

Count the overlap

CSRNet, CVPR 2018

Bayesian Loss, ICCV 2019

DM-Count, NeurIPS 2020

P2PNet, ICCV 2021

STEERER / APGCC, 2023-24

Tests: ShanghaiTech, QNRF, NWPU

Ground and currents

Chan perspective maps, CVPR 2008

Zhang ground-plane fusion, CVPR 2019

PACNN perspective, CVPR 2019

Mehran social force, CVPR 2009

Mehran streaklines, ECCV 2010

DIS optical flow (OpenCV)

Pressure and risk

Helbing social force, 1995

Helbing crowd disasters, 2007

Johansson pressure, 2008

Hughes continuum, 2002

Nakayama divergence, BMVC 2025

Harding early warning, PLoS ONE 2011

Status: idea stage. Numbers above are published research we build on, not our measurements.
Build from: DM-Count code (MIT) CSRNet model homography calibrator DeepStream docs

Disasters, limits, footage

Itaewon and Love Parade

159 dead, Seoul 2022, ignored calls. 21 dead, Duisburg 2010. Turbulence, not panic.

Three Indian disasters

Kumbh, Jan 2025: 30 dead, AI live. Hathras, Jul 2024: 121 dead, 80k vs 250k. Mumbai, Sep 2017: 23 dead, station footbridge.
Sources: Indian Express, Feb 2025; AP, Jul 2024; BBC, Sep 2017.

Danger thresholds

Fluid at 7 per m2. Serious above 4. Flow peaks at 2 to 3. (Fruin, Still)

Test footage

Public clips: station, crossing, temple.
Wikimedia Commons